Rameswar Panda

Distinguished Engineer, IBM Research

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Summary

AI research leader who builds open, efficient language models and agents that enterprises can trust and deploy. I lead the core IBM Granite language model team, 50+ researchers, engineers, and managers across the US and India. My team owns the full model lifecycle (architecture, pretraining, post-training, agentic reinforcement learning, reasoning and tool use, and evaluation). Principal investigator on multiple projects at the MIT-IBM Computing Research Lab, with 90+ papers (including NeurIPS, ICML, ICLR, CVPR, ICCV, and ECCV), 12,000+ citations, and 14 patents.

Selected Impact

- **About 10 major Granite releases**, from Granite Code through Granite 4.2, all under Apache 2.0, with 45M+ downloads on Hugging Face and Ollama and a record 95/100 on Stanford's Foundation Model Transparency Index.
- **IBM product impact:** Granite models power IBM products including IBM Bob, watsonx Code Assistant for Z, Db2, and Knowledge Catalog, and run on IBM Z and Power systems with the Spyre AI accelerator.
- **Architecture and efficiency:** hybrid Mamba-2/Transformer MoE in Granite 4.0, the first open models certified under ISO 42001; Granite 4.1 8B matched the previous 32B MoE.
- **Agentic AI:** built the agentic RL infrastructure and environments (SWE, terminal, search) behind Granite 4.2; released Toucan-1.5M, an open tool-use dataset used by NVIDIA for Nemotron.

Experience

IBM Research Cambridge, MA | Dec 2018 – Present

Distinguished Engineer, Granite Language Models & Agentic AI Oct 2024 – Present

Lead the core Granite team building IBM Granite language models and agentic AI systems, at IBM Research and the MIT-IBM Computing Research Lab.

- **Strategy and roadmap:** set the technical roadmap for the core Granite model team, from Granite 3.0 through 4.2.
- **Architecture:** co-designed Granite 4.0's hybrid Mamba-2/Transformer MoE and the dense Granite 4.1.
- **Reasoning and agentic RL:** built the RL behind Granite 3.2 reasoning and Granite 4.2's multi-environment agentic RL (SWE, terminal, and search agents); PRISM showed mid-training lifts RL gains 3–4$\times$.
- **Data:** created Toucan-1.5M, the #1 trending dataset on Hugging Face, and CodeAlchemy, a near-trillion-token synthetic code corpus used to train Granite.
- **Open source and ecosystem:** run model releases on watsonx, Hugging Face, Ollama, and LM Studio; Granite architectures are supported natively in Hugging Face Transformers, vLLM, and SGLang.
- **Cross-functional:** partner with IBM Software, Infrastructure, Consulting, and Red Hat to bring Granite into their products, and serve as principal investigator on multiple MIT-IBM projects on efficient LLM architectures.
- **People:** grew the team from 6 to 50+ people across the US and India, developing 6 first-line managers.

Principal Research Scientist Oct 2023 – Oct 2024

- Helped drive IBM Research's shift into open LLMs as corresponding author and technical lead of Granite Code Models (3B–34B, 116 languages, Apache 2.0), later extended to 128K context.
- Developed the Power Scheduler, a learning-rate schedule used in Granite pretraining that does not need retuning across batch sizes or token counts.
- The 20B Granite code model enabled IBM's first generative AI product for COBOL-to-Java modernization, watsonx Code Assistant for Z, and also powers Red Hat Ansible Lightspeed.

Research Scientist, MIT-IBM Watson AI Lab Dec 2018 – Sep 2023

- Led the team that built Granite 13B (granite.13b.instruct and granite.13b.chat), IBM's first in-house foundation models on watsonx.ai (2023).
- Drove multiple efficient AI and vision research projects in vision transformers, adaptive computation for video, and multi-task learning, including CrossViT (3,300+ citations) and a WACV Best Paper Honorable Mention.

University of California, Riverside Sep 2014 – Dec 2018

Graduate Student Researcher, Video Computing Group

Research internships: NEC Labs America (2018), Adobe Research (2017), Siemens Corporate Research (2016).

Education

Ph.D., Electrical and Computer Engineering, University of California, Riverside 2018
Advisor: Amit K. Roy-Chowdhury. Thesis: Visual Learning with Weak Supervision: Applications in Video Summarization and Person Re-identification.

M.S., Computer Engineering, Jadavpur University, India 2013
B.Tech., Electronics and Telecommunication Engineering, Biju Patnaik University of Technology, India 2011

Open Source

Granite models	Open models from 350M to 34B parameters under Apache 2.0, with 45M+ downloads.
Toucan-1.5M	1.5M tool-use trajectories from 495 real MCP servers; used to train many open-source models, including NVIDIA Nemotron and MiroThinker.
API Pack	1M+ instruction–API call pairs; used by Amazon (Hephaestus) and in Granite 3.0.
CodeAlchemy	Nearly 1T tokens of synthetic code across 15 languages; a 3B model trained on 100B synthetic tokens beat one trained on 600B real ones.
ScatterMoE, CrossViT	Triton MoE kernels used to train Granite MoE models; official CrossViT implementation.

Honors and Awards

- IBM: Special Accomplishment for watsonx (2023) • O-level Research Accomplishment (2022)
- Best Paper Honorable Mention, WACV (2023) • CVPR Doctoral Consortium Award (2018)
- S. Sue Johnson Endowed Graduate Award, UCR (2018) • Dean's Distinguished Fellowship, UCR (2014)
- Press: Granite covered by VentureBeat, ZDNET, The New Stack, and Interconnects; research covered by MIT News.

Professional Service

- **Area Chair:** NeurIPS 2026, ICML 2027, AAAI 2027, WACV 2022, ICVGIP 2022.
- **Workshop organizer:** Dynamic Neural Networks (CVPR 2021, 2022), Neural Architecture Search (CVPR 2020, 2021), Multi-Modal Video Analysis (ICCV 2019, ECCV 2020).
- **Tutorials:** Efficient Video Understanding (ICCV 2021), Visual Data Summarization (CVPR 2019).
- **Program committee:** NeurIPS, ICML, ICLR, AAAI, CVPR, ICCV, ECCV.

Selected Publications

90+ papers, 12,000+ citations, h-index 49. Full list on Google Scholar.

[1] TOUCAN: Synthesizing 1.5M Tool-Agentic Data from Real-World MCP Environments. Z. Xu, A. Meza Soria, S. Tan, A. Roy, A. Agrawal, R. Poovendran, **R. Panda**. *EMNLP 2026*.

[2] Dynamic Short Convolutions Improve Transformers. O. Sieberling, B. Runwal, **R. Panda**, Y. Kim. *NeurIPS 2026*.

[3] Distilling to Hybrid Attention Models via KL-Guided Layer Selection. Y. Li, S. Yang, S. Tan, M. Mishra, **R. Panda**, J. Zhou, Y. Kim. *ICLR 2026*.

[4] CodeAlchemy: Synthetic Code Rewriting at Scale. A. Gupta, A. Prasad, **R. Panda**. *arXiv 2026*.

[5] PRISM: Demystifying Retention and Interaction in Mid-Training. B. Runwal, A. Agrawal, A. Roy, **R. Panda**. *arXiv 2026*.

[6] PaTH Attention: Position Encoding via Accumulating Householder Transformations. S. Yang, Y. Shen, K. Wen, S. Tan, M. Mishra, L. Ren, **R. Panda**, Y. Kim. *NeurIPS 2025*.

[7] Scaling Stick-Breaking Attention: An Efficient Implementation and In-depth Study. S. Tan, S. Yang, A. Courville, **R. Panda**, Y. Shen. *ICLR 2025*.

[8] Reducing Transformer Key-Value Cache Size with Cross-Layer Attention. W. Brandon, M. Mishra, A. Nrusimha, **R. Panda**, J. Ragan-Kelley. *NeurIPS 2024*.

[9] Gated Linear Attention Transformers with Hardware-Efficient Training. S. Yang, B. Wang, Y. Shen, **R. Panda**, Y. Kim. *ICML 2024*.

[10] Granite 3.0 Language Models. IBM Granite Team (technical leadership). *Technical report 2024*.

[11] Granite Code Models: A Family of Open Foundation Models for Code Intelligence. M. Mishra, M. Stallone, G. Zhang, et al., D. Cox, R. Puri, **R. Panda**. *arXiv 2024*.

[12] Select, Label, and Mix: Learning Discriminative Invariant Feature Representations for Partial Domain Adaptation. A. Sahoo, **R. Panda**, R. Feris, K. Saenko, A. Das. *WACV 2023, Best Paper Honorable Mention*.

[13] CrossViT: Cross-Attention Multi-Scale Vision Transformer for Image Classification. C.-F. Chen, Q. Fan, **R. Panda**. *ICCV 2021*.